

Submission PHYS1-FINAL-SUB08

Question PHYS1-FINAL-Q01

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visible work:

$$v = v_i + at = 8 + 2(5) = 18 \text{ m/s} \dots ?$$

last line is hard to read; maybe $v = 18 \text{ m/s}$; displacement = 65 m

Question PHYS1-FINAL-Q02

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visible work:

$$N = mg = 19.6 \text{ N}; \text{ friction} = \mu N = 3.9 \text{ N} \dots ?$$

last line is hard to read; maybe $a = 4.54 \text{ m/s}^2$; friction opposes motion

Question PHYS1-FINAL-Q03

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visible work:

$$mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{(2gh)} = \sqrt{(2 \cdot 9.8 \cdot 6)} \dots ?$$

last line is hard to read; maybe $v = 10.84 \text{ m/s}$

Question PHYS1-FINAL-Q04

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visible work:

$$p_{\text{total, before}} = p_{\text{total, after}} \dots ?$$

last line is hard to read; maybe total momentum before equals after when external impulse is negligible

Question PHYS1-FINAL-Q05

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visible work:

$$\text{Impulse is area under } F(t): J = F\Delta t = 6(0.5) = 3 \text{ N}\cdot\text{s} \dots ?$$

last line is hard to read; maybe impulse = 3 N·s; momentum changes by 3 kg·m/s

Question PHYS1-FINAL-Q06

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visible work:

$$v = v_i + at = 5 + 2(5) = 15 \text{ m/s} \dots ?$$

last line is hard to read; maybe $v = 15 \text{ m/s}$; displacement = 50 m

Question PHYS1-FINAL-Q07

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visible work:

$$N = mg = 68.6 \text{ N}; \text{ friction} = \mu N = 13.7 \text{ N} \dots ?$$

last line is hard to read; maybe $a = 1.33 \text{ m/s}^2$; friction opposes motion

Question PHYS1-FINAL-Q08

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visible work:

$$mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2 \cdot 9.8 \cdot 5} \dots ?$$

last line is hard to read; maybe $v = 9.90 \text{ m/s}$

Question PHYS1-FINAL-Q09

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visible work:

$$p_{\text{total, before}} = p_{\text{total, after}} \dots ?$$

last line is hard to read; maybe total momentum before equals after when external impulse is negligible

Question PHYS1-FINAL-Q10

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visible work:

$$\text{Impulse is area under } F(t): J = F\Delta t = 6(0.5) = 3 \text{ N}\cdot\text{s} \dots ?$$

last line is hard to read; maybe impulse = $3 \text{ N}\cdot\text{s}$; momentum changes by $3 \text{ kg}\cdot\text{m/s}$